Water Bath



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Water Bath

What it is used for

A laboratory water bath maintains samples at a controlled temperature using heated water. It is used for incubation, warming, and temperature-dependent procedures that require uniform heating.

Main parts

- Insulated bath tank
- Heating element
- Temperature sensor and controller
- Sample racks or holders
- Lid or cover
- Drain outlet and power switch

Before you start

1. Read the operating instructions and confirm the required temperature range for the procedure.
2. Inspect the tank, power cable, controller, sensor, lid, and drain for damage.
3. Place the water bath on a stable, level, dry surface away from splashes.
4. Fill the tank with the recommended water to the specified operating level.
5. Check that the heating element is covered to the required level before switching on.
6. Verify the temperature sensor and any required calibration status.

How to use

1. Switch on the water bath and set the target temperature using the controller.
2. Allow the water to heat and stabilize at the set temperature.
3. Check the displayed temperature against an independent thermometer when required by the procedure.
4. Place samples in suitable sealed or approved containers and arrange them in the rack.
5. Lower the rack carefully so water does not enter sample containers.
6. Ensure the water level remains within the operating range during use.
7. Monitor the temperature and sample exposure time throughout the procedure.
8. Remove samples using suitable tools or heat-resistant gloves when the procedure is complete.
9. Switch off the heater and allow the water to cool before draining or cleaning.
10. Drain, clean, and dry the tank according to the laboratory procedure.

⚠ Safety

- Do not operate the bath without sufficient water or with an exposed heating element unless designed for it.
- Avoid touching hot water, tank surfaces, or sample containers with bare hands.
- Keep electrical connections and the control panel dry.
- Use suitable racks and containers that tolerate the operating temperature.
- Disconnect power before draining, cleaning, or servicing the equipment.
- Do not use the bath for flammable or hazardous liquids unless specifically approved by the manufacturer and laboratory procedure.

Care & storage

- Replace bath water at the frequency required by the laboratory procedure.
- Clean the tank and remove scale using approved materials.
- Inspect the sensor, heater, cable, and controller for damage or abnormal operation.
- Record temperature checks, calibration, cleaning, and maintenance as required.

Common mistakes to avoid

- Operating with too little water can overheat and damage the heater.
- Overfilling the tank can cause spills when racks or samples are inserted.
- Using an unsealed sample container can allow water to contaminate the sample.
- Ignoring temperature drift can invalidate temperature-sensitive test results.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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